

OpenClaw Temperature and Humidity Dashboard Tutorial

Before starting, please complete [01-Peripheral hardware configuration], [02-openclaw API-KEY configuration], [03-openclaw switching model], [04-OpenClaw prompt words], [05-AI voice interaction configuration], [06-Three dialogue scheme configuration tests] in the `Preparation before use` directory. If you are currently only using TUI/WhatsApp, you can skip `05` first. By default, the basic environment of this article has been prepared and only the temperature and humidity dashboard project itself will be expanded.

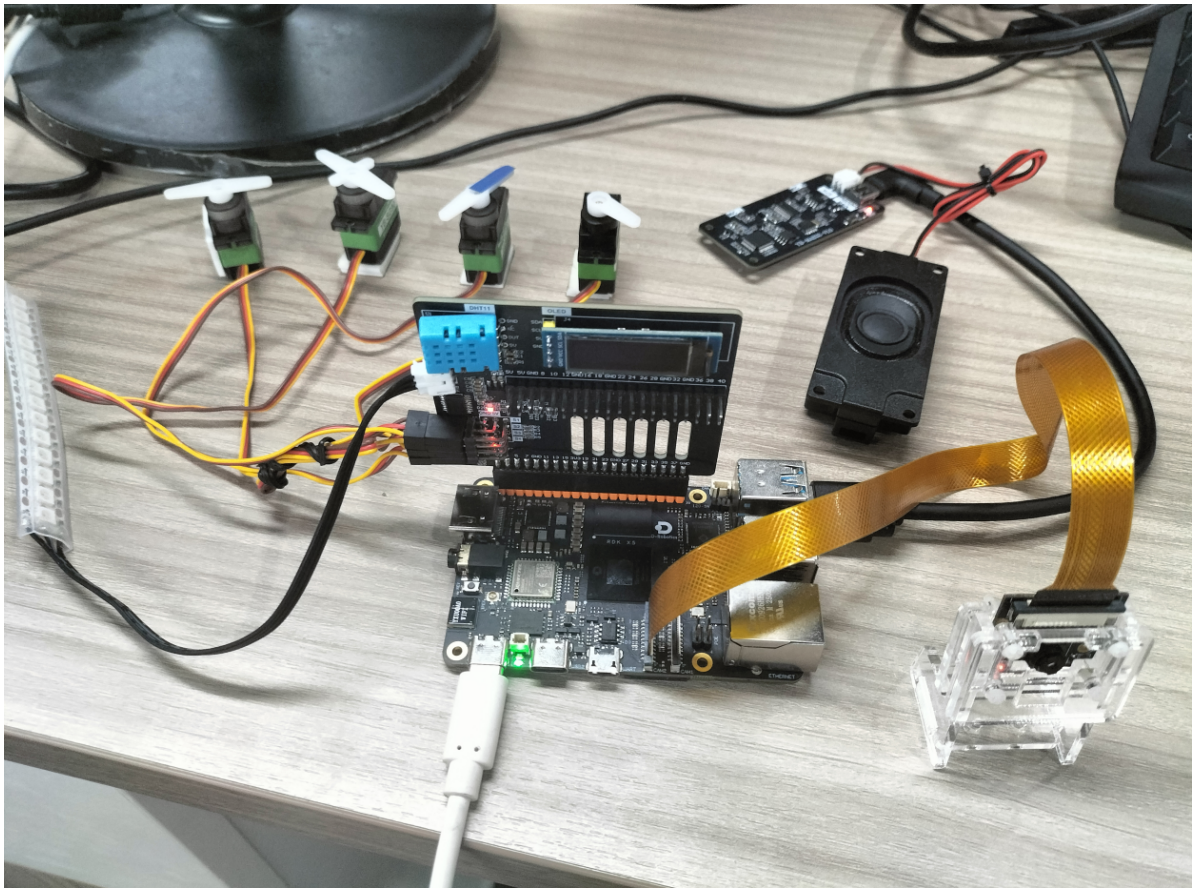
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral collaboration capabilities in a desktop dashboard scenario through a combination of "sensor + display + indication". After completing this tutorial, readers will be able to:

- Learn how to combine DHT, OLED, RGB and servo into a temperature and humidity dashboard
- Converse with OpenClaw through WhatsApp, TUI, and voice code
- Complete four project scenarios: real-time reading, OLED indicator panel, RGB status light, and servo pointer mapping

2. Hardware preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip/RGB lamp bead module
- `[optional] AI voice module + speaker`
- Wiring diagram



3. Software preparation

Code entry that will be used in the tutorial

- Temperature and humidity reading command: `/home/sunrise/hardware_ha1_demo/dht`
- OLED control command: `/home/sunrise/hardware_ha1_demo/oled`
- RGB control command: `/home/sunrise/hardware_ha1_demo/rgb`
- Servo control command: `/home/sunrise/hardware_ha1_demo/cli/servo`
- Voice entrance: `/home/sunrise/voice_to_openc1aw/src/main.py`

This project is very close to the AI weather station, but the focus is different:

- AI weather station places more emphasis on "broadcast" and "explanation"
- The temperature and humidity dashboard places more emphasis on "display" and "status mapping"

4. Test the basic capabilities of the project

Before starting the current project, please complete a basic dialogue self-test of `openc1aw tui`; if you have not done so yet, please read [05-Three dialogue scheme configuration tests] first. After passing, enter the special test of temperature and humidity instrument panel.

It is recommended to enter OpenClaw TUI first:

```
openc1aw tui
```

```
sunrise@ubuntu:~/openclaw$ openclaw tui
🔔 OpenClaw 2026.3.24 (cff6dc9) - You had me at 'openclaw gateway start.'
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

hello

你好呀，主人！👋 我是小亚，随时待命。
有什么需要我帮忙的吗？
gateway connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 12k/1.0m (1%)
```

Then test the following types of instructions in sequence:

- Read current temperature and humidity
- Display current temperature and humidity on OLED
- If the environment is normal, the green light will turn on
- Turn the servo to 90 degrees

5. Three options for talking to OpenClaw

Solution	Suitable scenario	Advantages	Suggested positioning
WhatsApp	Remote status viewing, classroom demonstration	Convenient remote reading of current panel data	Display of "Remote Monitoring Panel"
TUI	Local debugging, mapping verification	The easiest way to check the relationship between readings and actions	Run through the visual link first
Voice code	Voice broadcast, desktop assistant	Closer to daily use	As a complete demonstration solution

If the AI voice module + speaker is optional, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to open broadcast parameters

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter is to control whether to broadcast. After modification, save it and run the following program on the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp solution

For WhatsApp access and robot creation steps, please refer directly to [06-Configuration Test of Three Dialogue Schemes]. After creating it, you can remotely view the status of the dashboard in WhatsApp, for example:

- Temperature and humidity dashboard

20:36



< 2



+86 153 3885 7526 (你)

给自己发消息

warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓

Temperature and humidity dashboard

20:35 ✓✓

Temperature and humidity dashboard is now active!



Current Readings:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 54%



OLED Display: Showing temp and humidity values

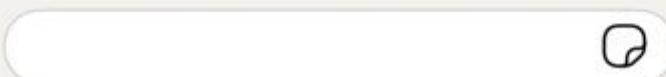


RGB Indicator: Red blinking light (speed 5)

- This indicates the temperature is **above 28°C** (warm/hot zone)

Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!

20:35 ✓✓



5.2 TUI scheme

If you have completed the TUI self-test in Part 6, enter the terminal text dialogue directly:

```
openclaw tui
```

Then enter project-related instructions, such as:

- Temperature and humidity dashboard

```
File Edit Tabs Help
The weather card is now showing on your OLED screen! ✓

、 Temperature and humidity dashboard

Temperature dashboard activated! 🌡️🌡️

Current Readings:
- Temperature: 30.0°C (86°F)
- Humidity: 52%

Actions:
- ✓ OLED: Displaying "Temp: 30.0C, Humidity: 52%"
- ✓ RGB: Red blinking effect (speed 5) - indicates high temperature (>28°C)

Status: ⚠️ Temperature is above the comfortable range (28°C threshold). The
red blinking light serves as a heat warning!

The dashboard is now running and will continue to show the current
environmental conditions. 🌡️🌡️
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
82k/1.0m (8%)
```

5.3 Voice code scheme

Requires optional ai voice interaction module + speaker

For voice module installation, environment configuration and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After waking up, you can directly say the control content, such as:

- Temperature and humidity dashboard

```

[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Temperature and humidity dashboard.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 🌡️🌡️Temperature & Humidity Dashboard Active!

**Current
  Readings:**
- **Temperature:** 30.1°C (86.2°F)
- **Humidity:** 51%
  **Dashboard Status:**
- ✔️ OLED displaying real-time values
- ✔️ RGB
  indicator: **Red blinking** (heat warning - above 28°C)
  **Temperature Color Code:**
- 🌡️ Blue breathing: < 20°C (Cold)
- 🌡️
  Solid green: 20-28°C (Comfortable)
- 🌡️ Red blinking: > 28°C (Hot) ⚠️
  **Current Status:** **HOT** - Temperature is above the comfort
  threshold! Consider cooling measures. 🌡️🌡️

The dashboard will continue
  showing live readings. Want me to check again in a few minutes?

```

6. Project implementation

Note: The following are all dialogue demonstrations in `openclaw tui` mode. You can also choose WhatsApp or voice mode to complete it.

6.1 Scenario 1: Real-time reading

Experimental goals

Read the current temperature and humidity and complete the basic data input of the dashboard.

Effect description

OpenClaw will first get the temperature and humidity readings, and these values are the basis for subsequent OLED, RGB and servo mapping.

Example instructions

- Read current temperature and humidity
- What is the current temperature and humidity?
- Help me refresh the current readings

6.2 Scenario 2: OLED indicator disk

Experimental goals

Display the current temperature and humidity results into a concise OLED dashboard content.

Effect description

It is suitable to use the `metrics` or `dht_oled` method to combine `Temp` and `Hum` into a compact screen area to form a small data panel.

Example instructions

- Display Temp and Hum on OLED
- Make a simple temperature and humidity panel
- Write the current temperature and humidity on the screen

6.3 Scene 3: RGB status light

Experimental goals

Use RGB colors to express whether the current environment is normal, hot, cold, or dry.

Effect description

This link can make the dashboard more intuitive. For example, normal bright green, hot bright yellow, abnormal bright red, cold bright blue, the status can be seen at a glance.

Example instructions

- If the environment is normal, the green light will turn on
- If the temperature is high, the yellow light will turn on
- If the humidity is abnormal, the red light will turn on

6.4 Scenario 4: Servo pointer mapping

Experimental goals

Map temperature or humidity to servo angle so that the servo points to a position like a dashboard needle.

Effect description

For example, if the temperature range is mapped to 0 to 180 degrees, the higher the current temperature, the further to the right the servo pointer will be. This effect is very suitable for classroom demonstrations, because "numbers turn into physical actions" will be very intuitive.

Example instructions

- Use the servo to indicate the current temperature like a pointer
- Map current temperature to servo angle
- Let the pointer point to the current state